

1.] output following

```
S = "Python"
```

```
Print(S[0], S[-1], S[6])
```

Ans S[0] = P

S[-1] = n

S[6] = Index out of Range

2.] Positive Indexing

* Positive Indexing start from left

* And Begin Index value 0

Negative Indexing

* Negative Indexing start from Right side.

* And begins with Index value of [-1]

```
S = "Hello"
```

```
S[-2] ⇒ [l] Ans.
```

3.] * Index out of Range → when we access Index in out of Range in String Python produce [Index error]

* Slicing → But in slicing out of Range of string does not given an error. It returns The until The Range.

```
S = "code"
```

```
print(S[10]) # Index error - Out of Range.
```

Print (s[1:10]) # ode

7] To check The first and last occurrence of a character using The value of Index [eg s[2]].

First $\rightarrow$ Index 0 to length-1

last $\rightarrow$ Index length-1 to 0.

S = "mississippi"
0 1 2 3 4 5 6 7 8 9 10

first 's' # s[2]

last 's' # s[6]

5] Negative Indexing

S = "abcdef"

a b c d e f
-6 -5 -4 -3 -2 -1

Print (s[-1], s[-3], s[-6])

#s[-1] $\rightarrow$ f #s[-3] $\rightarrow$ d #s[-6] $\rightarrow$ a

6]

Syntax [s[start:stop:step]]

Start $\rightarrow$ Starting Index value [eg s(1)]

Stop $\rightarrow$ Ending Index value [eg s[10]]

Step $\rightarrow$ Number of position to move.

7] `s = "Hello world"`

 Hello world
 0 1 2 3 4 5 6 7 8 9

~~Print~~ `Print(s[0:5])` # start from [0] ends in [4] because length-1

Ans → Hello

`Print(s[5:0])` # Ending range not given so print all.

Ans → world

`Print(s[:5])` # start Index is not given always considered as [0]

Ans → Hello

`Print(s[:])` # not indexing value give

Ans → Hello world.

8] If start value is greater than stop value in positive step. Python returns the empty string.

`s = "Python"`

`Print(s[4:1])` # ' '

9.) We extract last 3 digit of character in string using negative slicing method.

`s = "Python"`

`Print(s[-3:])` # 'ion'

s = "a b c d e f g h"

Print (s[::1]) $\Rightarrow$ # Start Index [0] Step Increment [1]
a b c d e f g h

Print (s[1::2])
~~b d f h~~ a c e g

Print s [1::2] # starting Index [1] step Increment [2]
b d f h

] Reverse a string using slicing

Start $\Rightarrow$ default

Stop $\Rightarrow$ default

Step $\Rightarrow$ [-1] move from backwards.

S = "Python"

Print (s[::-1])

S = nohtyP

S = a b c d e f g h

Print (s[::-1]) # h g f e d c b a

Print (s[::2]) # h f d b

Print (s[6:1:-1]) # g f e d c

3] `s = "a1b2c3d4e5"`

`Print(s[1::2])`

1 2 3 4 5

4] `s = "Python"`

`Print(s[s:0:-1])` # Start [s] end [0] step Increment [1]

↳ not typ

`Print(s[s::-1])` # start [s], end default, step [1]

↳ not typ

`Print(s[:0:-1])`

↳ not typ

15.]

In a negative step Index move from Right → left

`[start(last Index) : stop [before first Index) : step]`

`s = "hello"`

`Print(s[100::-1])` # last Index 100 But python

↳ olleh valid The last Index [s]

`Print(s[::-100])` # start [default] stop [default]

↳ 0

Step [-100] Out range.

16.]

`s = "abcde fghij "`

`Print(s[2:-1])`

start [2] value and stop before last value.

palindrome

s = "racecar"

If s == (s[::-1])

Print ("Palindrome")

else

Print ("not a palindrome")

s = "Hello"

[Hint why not raise error]

Print (s[1:100]) start [1] end [100] out range but python takes valid index.
↳ ello

Print (s[-100:3]) start [-100] out of range python adjust to start [0]
↳ Hello

s = hello world

Print = s[:6] + "Python"

Ans → hello Python

Im python string cannot replace .so take the specific string using index. Slicing and concatenation.

s = "abcdefghij"

my-slice = slice(1,8,2)

Print (my-slice)

↳ bdfgh

slice[1:8:2]

a b c d e f g h i j
0 1 2 3 4 5 6 7 8 9